AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M348
Applied Statistical Modelling

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6 /Dec. /2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: 30
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: True or False Questions [10x 0.5=5 marks]

Which of the following statements are true/false?

a) If TSS = ESS, it means the regression model perfectly fits the data.
b) In simple linear regression, β_1 represents the change in Y for a one-unit change in X.
c) The F-test in regression checks whether all regression coefficients are simultaneously zero.
d) A coefficient equal to zero still contributes to the prediction of Y.
e) Multicollinearity occurs when predictors are highly correlated.
f) Backward selection begins with all variables and removes the least significant ones.
g) AIC is maximized, not minimized, during model selection.
h) Polynomial regression always outperforms linear regression, even if the true relationship is linear.
i) For right-skewed data, we move up the ladder of powers (e.g., square, cube).
j) Transformations are sometimes necessary to meet regression assumptions.

PART 2: Multiple Regression for the Energy Efficiency dataset [16 marks]

ALL QUESTIONS ARE REQUIRED

This dataset contains building characteristics and their associated heating and cooling loads. It is commonly used for regression analysis in predicting building energy efficiency. The dataset includes 768 records of simulated buildings, each described by 8 features and 2 target variables.

Variables:

- Relative_Compactness : Ratio of building volume to its envelope area (a measure of compactness).
- Surface_Area :Total surface area of the building in square meters.
- Wall_Area :Total area of the building walls in square meters.
- Roof_Area :Total area of the building roof in square meters.
- Overall_Height :Total height of the building in meters.
- Orientation: Orientation of the building (encoded as integer categories).
- Glazing_Area: Total window/glazing area as a fraction of the floor area.
- Glazing_Area_Distribution : Distribution of windows/glazing across the building façades (encoded as categories 0–5).
- Heating_Load : Heating load requirement of the building (target variable 1).
- Cooling_Load : Cooling load requirement of the building (target variable 2).

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In this part, we will fit a multiple regression model for the response variable Heating_Load from the Energy Efficiency dataset, using Relative_Compactness, Surface_Area, Wall_Area, Overall_Height and Glazing_Area as explanotary variables.

```
Residuals:
    Min      1Q  Median      3Q     Max
-10.3862 -1.3667 -0.0142  1.3162  7.5555

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)       84.386471  19.111765   4.415 1.15e-05 ***
Relative_Compactness -64.773432  10.333611  -6.268 6.11e-10 ***
Surface_Area       -0.087289   0.017149  -5.090 4.51e-07 ***
Wall_Area          0.060813   0.006676   9.109  < 2e-16 ***
Overall_Height     4.169954   0.339441  12.285  < 2e-16 ***
Glazing_Area      20.437968   0.798727  25.588  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.947 on 762 degrees of freedom
Multiple R-squared:  0.9153,    Adjusted R-squared:  0.9147
F-statistic:  1646 on 5 and 762 DF,  p-value: < 2.2e-16
```

1) Why is multiple regression more beneficial than using a simple regression with only Relative_Compactness as the predictor for Heating_Load?
2) Interpret the coefficient of Glazing_Area (e.g., 20.44) in the model. What does this value mean in the context of heating load?
3) What does the adjusted $R^2 = 0.9147$ tell us about the model, and why is adjusted R^2 more reliable than multiple R^2 when comparing models?
4) Which explanatory variable has the strongest positive effect on Heating_Load and which has the strongest negative effect? Base your answer on coefficients and p-values.
5) The residual standard error (RSE) is 2.95. Suppose a previous simple regression model (with only Overall_Height as predictor) had an RSE of 9.3. What does this reduction suggest about the model's improvement?
6) How could multicollinearity among Surface_Area, Wall_Area, and Relative_Compactness affect the interpretation of coefficients? Suggest one method to detect it and one method to address it.
7) Suppose you add more predictors such as Roof_Area, Orientation, and Glazing_Area_Distribution. Write the R code to perform

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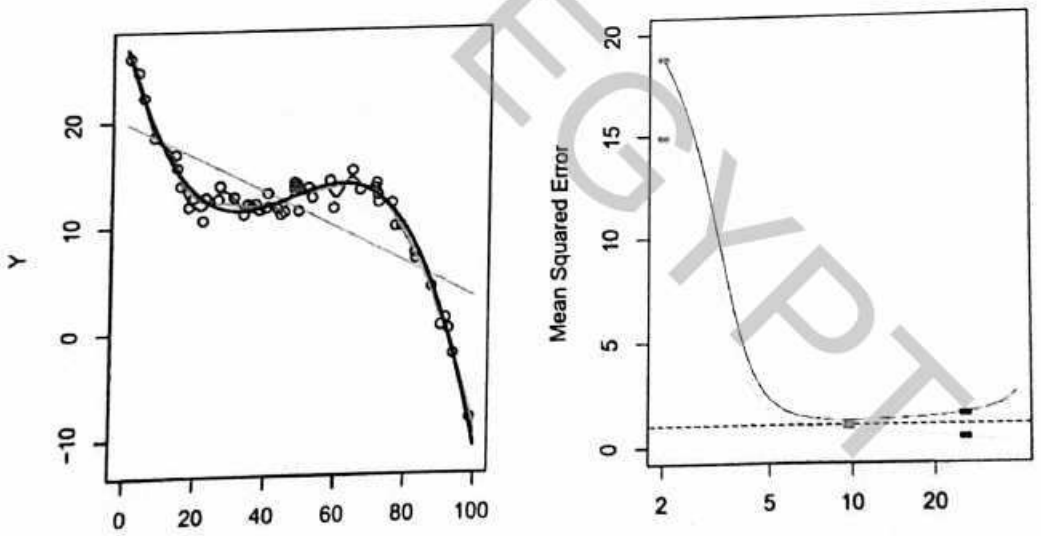
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forward starting from null model and backward stepwise regression starting from the full model.

8) What is the purpose of using stepwise regression in model selection?
9) When applying stepwise regression, should we minimize or maximize the AIC? Explain why.
10) Suppose the final stepwise model selected only Relative_Compactness, Overall_Height, and Glazing_Area. What does this result tell you about the importance of the excluded predictors (Surface_Area and Wall_Area)?
11) Using the fitted model, predict the Heating_Load of a building with the following characteristics:
    i. Relative_Compactness = 0.85
    ii. Surface_Area = 500
    iii. Wall_Area = 250
    iv. Overall_Height = 7
    v. Glazing_Area = 0.25
12) Suppose two competing models both have high adjusted R^2 but one has a much higher residual standard error. Which model would you choose, and why?

PART 3: Assessing Flexibility in Regression Methods [9 marks]

ALL QUESTIONS ARE REQUIRED



The figure below consists of two panels:

- **Left panel: Data simulated from a true nonlinear function f (black curve). Three models are fit:**
  - **Linear regression (orange line).**

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Two smoothing splines with different degrees of flexibility (blue and green curves).

Right panel: Plots of training mean squared error (MSE, grey curve, below), test MSE (red curve, above), and the minimum possible test MSE achievable by any method (dashed line). Colored squares correspond to the training and test MSEs for the three models shown in the left panel.

1. Model Flexibility:
   a) From the left panel, identify which model is the most flexible and which is the least flexible. Explain the criteria used to make this determination.
2. Overfitting:
   a) Based on the right panel, explain what evidence suggests that the green spline may be overfitting.
3. MSE Evaluation:
   a) Why is the test MSE (red curve) more important than the training MSE (grey curve) when assessing a model's performance?
4. Bias–Variance Tradeoff:
   a) Briefly define bias and variance in the context of statistical learning.
5. Parametric vs Non-Parametric:
   a) Classify the three models (orange line, blue spline, green spline) as either parametric or non-parametric.
   b) What are the key advantages and disadvantages of using a parametric approach (like the orange line) compared to a non-parametric one (like the splines)?
6. Applied Question (Critical Thinking):
   a) In what situations would a flexible non-parametric model be more appropriate than a simple linear regression?

End of questions

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